

Math 142, S12

Name: Answer

Quiz 5

Section:

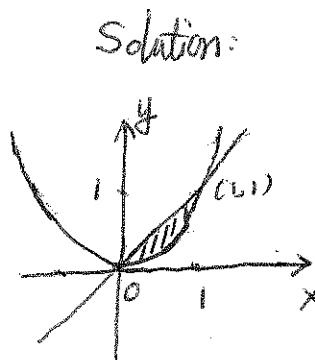
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Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (5 points) Evaluate the improper integral $\int_2^3 \frac{1}{\sqrt{x-2}} dx$ if possible.

$$\begin{aligned}
 \text{Solution: } \int_2^3 \frac{1}{\sqrt{x-2}} dx &= \lim_{t \rightarrow 2^+} \int_t^3 \frac{1}{\sqrt{x-2}} dx \\
 &= \lim_{t \rightarrow 2^+} (2\sqrt{x-2}) \Big|_t^3 \\
 &= \lim_{t \rightarrow 2^+} (2 - 2\sqrt{t-2}) \\
 &= 2
 \end{aligned}$$

Problem 2. (5 points) Sketch the region enclosed by $y = x$, $y = x^2$ and find its area.



Intersection points

$$\begin{aligned}
 \begin{cases} y=x \\ y=x^2 \end{cases} &\Rightarrow x=x^2 \\
 x(x-1) &= 0 \Rightarrow x=0, x=1 \\
 \downarrow &\quad \downarrow \\
 y=0 &\quad y=1
 \end{aligned}$$

Thus

$$\begin{aligned}
 A &= \int_0^1 (x - x^2) dx \\
 &= \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^1 \\
 &= \left(\frac{1}{2} - \frac{1}{3} \right) - 0 \\
 &= \frac{1}{6}
 \end{aligned}$$